

LAB #8: Setting Up VLAN (part 1)

Lab Requirements:

- Packet Tracer tool
- Lab Document
- Windows/Mac OS Host

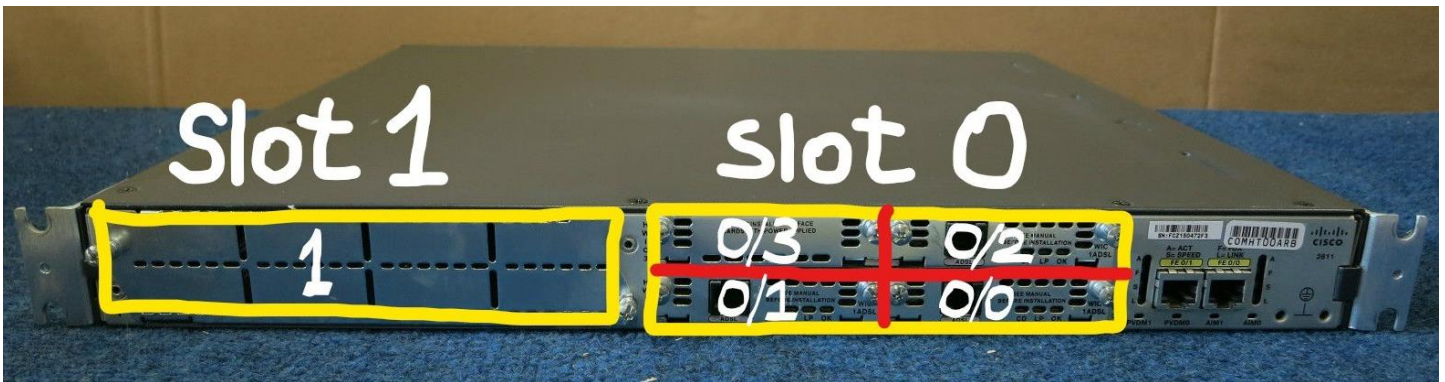
1. Overview

In this in lab, you will work on Cisco Packet Tracer (CPT) simulator. You are a member of a team working as networking specialists at Stanford University. One of the school buildings has a dedicated link to connect their Willowbrook campus to the SugarLand campus.



2. Router/IOS

You will use the Cisco 2811 Series router.



Cisco 2811 Router Rear View

Each network interface on a Cisco 2811 series router is identified by a slot number and a port number. A Cisco 2811 series router chassis contains two, four, or six slots in which you can install modules. You can install any module into any available slot in the chassis.

For the Cisco 2811 router, the slots are numbered as follows:

- Slot 0 is at the right with four separate sub-slots (numbered as 0/0, 0/1, 0/2, 0/3).
- Slot 1 is on the left.

Port numbers begin at 0 for each slot and continue from right to left and (if necessary) from bottom to top. Modules and WAN interface cards are identified by interface type, slot number, a forward slash (/), and the port number, for example Ethernet 0/0.

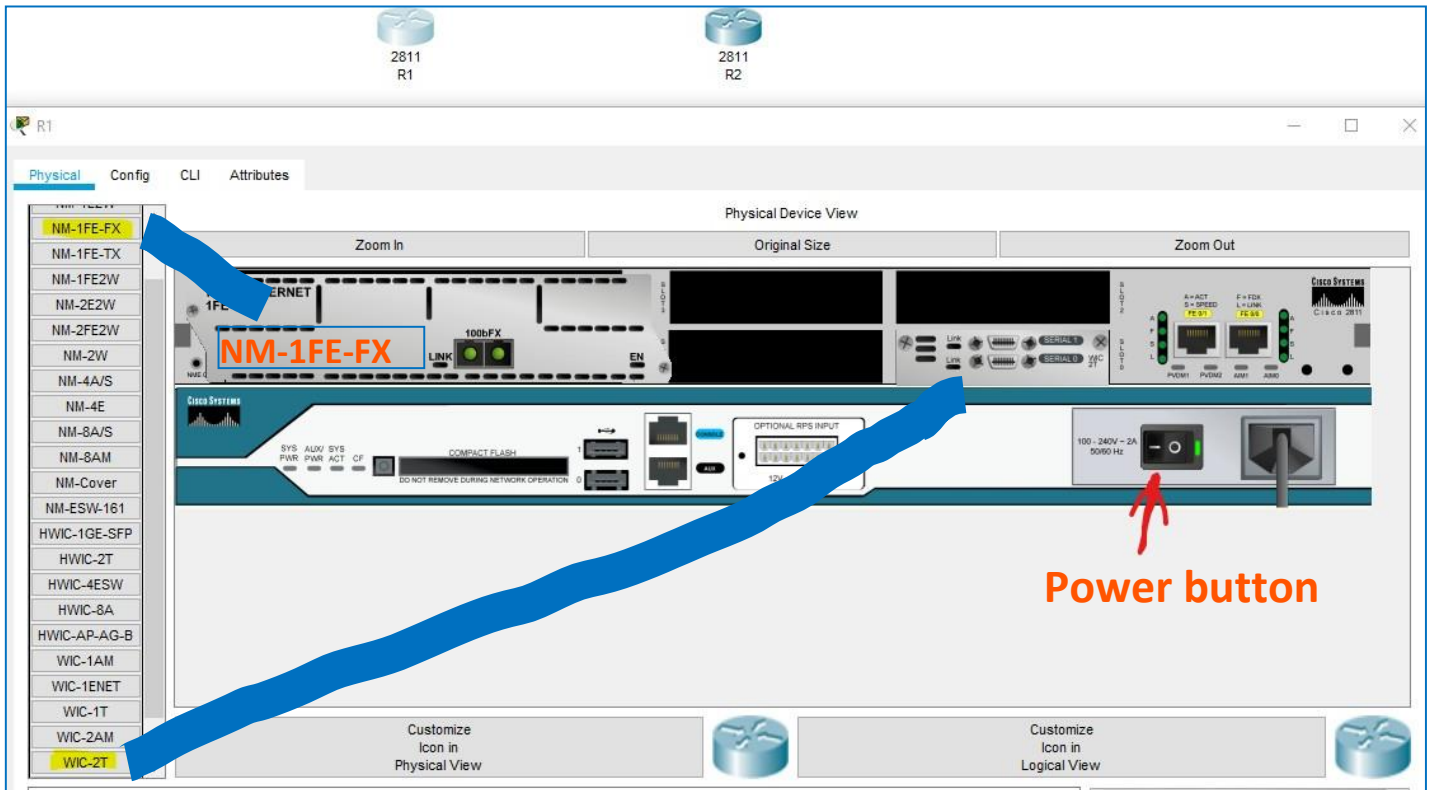
3. CPT Simulation Structure

We use functions to break a problem into smaller sub-problems. This reduces program complexity. The main function will be the driver of the program.

3.1 Topology

Let's build the topology shown with two routers. Start Cisco Packet Tracer.

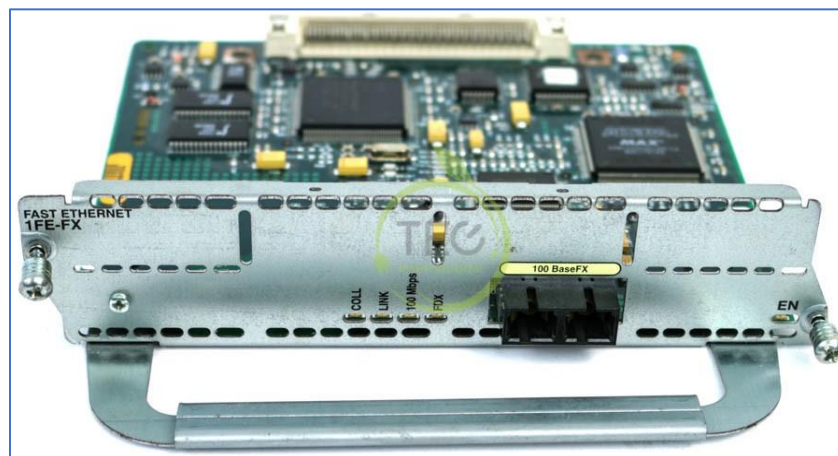
1. Drag two 2811 model routers into the workspace. CPT gives each router a display name beginning with Router0. To change display name, click on the display name of the router then change Router0 to R1 and Router1 to R2.
2. Click on R1 and then the Physical tab. Click on the power button to turn off the router then drag "NM-1FE-FX" module to the Slot1 (biggest slot on the left side of router) and "WIC-2T" module to the Slot0/0. We will use "NM-1FE-FX" for LAN fiber cable connection and "WIC-2T" for WAN Serial DTE cable connection. Now turn on the router by clicking power button.



This will add a NM-1FE-FX Fast Ethernet adapter to the router.

The NM-1FE-FX Module provides one Fast-Ethernet interface for use with fiber media. Ideal for a wide range of LAN applications, the Fast Ethernet network modules support many internetworking features and standards.

Single port network modules offer autosensing 100BaseFX Ethernet.



NM-1FE-FX Rear View

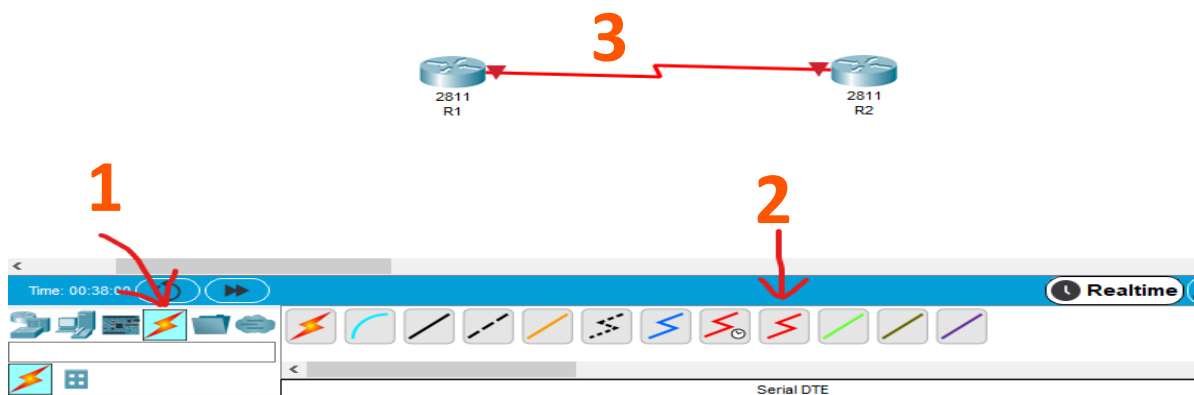
The WIC-2T dual-serial port WAN interface cards (WICs) feature Cisco's new, compact, high-density Smart Serial connector to support a wide variety of electrical interfaces when used with the appropriate transition cable. Each port on a WIC is a different physical interface and can support different protocols such as Point-to-Point protocol (PPP) or Frame Relay and Data Terminal Equipment/Data Communications Equipment (DTE/DCE).



WIC-2T Rear View

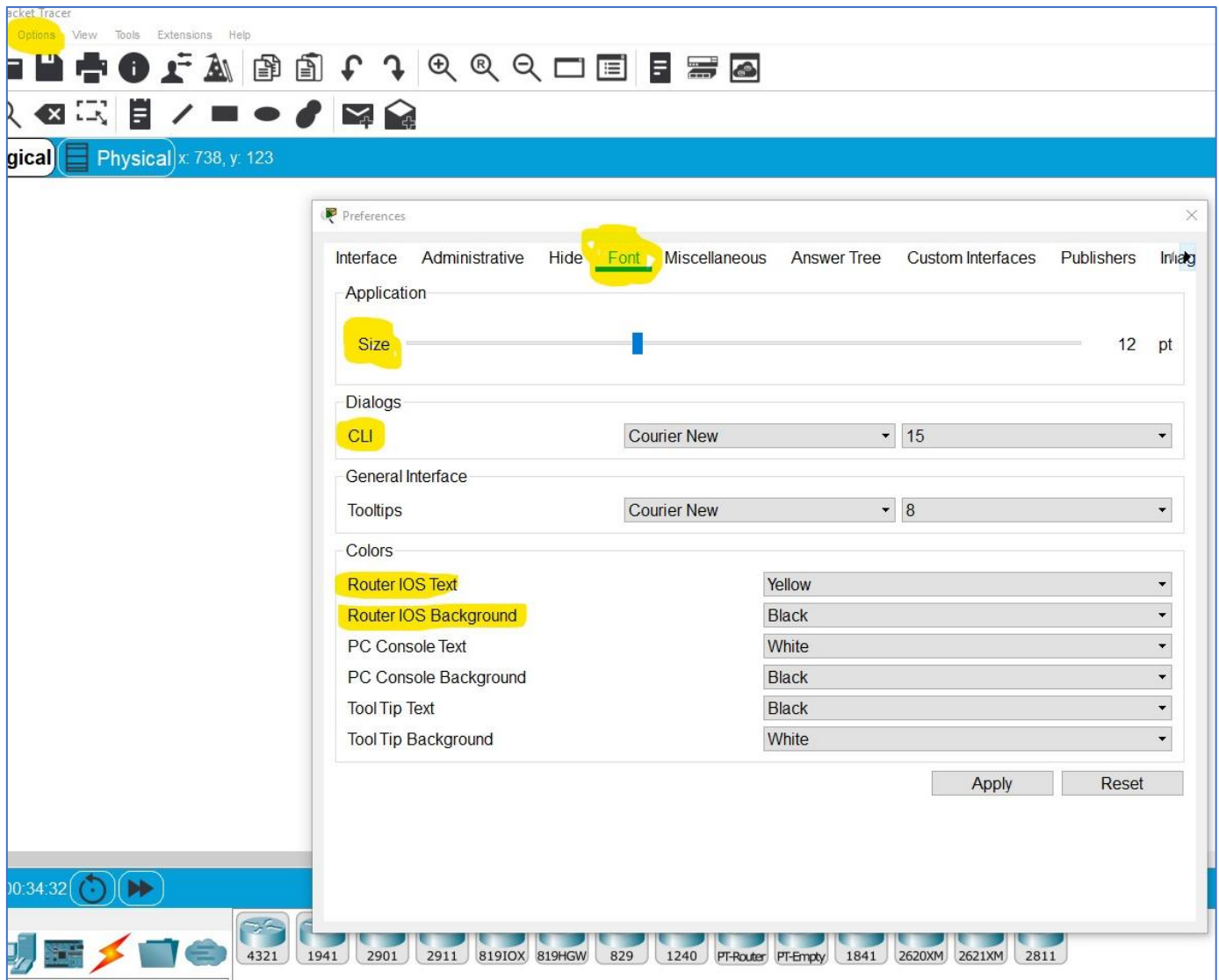
Add the same adapters to the second router. Now you are ready to connect the routers.

- 1) Click the Connections icon from bottom left
- 2) Select the Serial DTE
- 3) Click on R1 and select Serial0/0/0 then click R2 and select Serial0/0/0.



You may now configure each router. In our lab study, you will change the hostnames, assign IP addresses to the interfaces, and turn them on, and finally test connectivity by using ping.

Before proceeding we recommend changing CLI font size and color. Go to Options and click the Preferences then select Font tab. Update the default preferences as shown below with the yellow highlight.

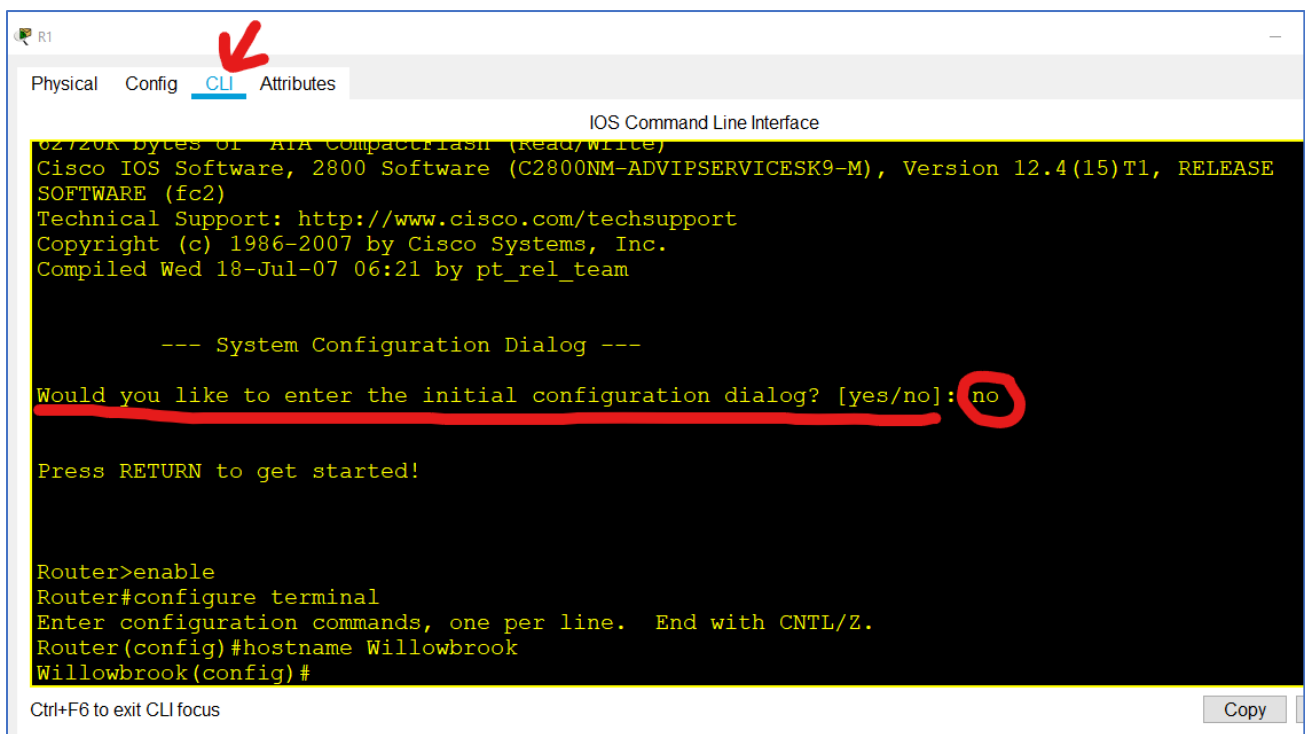


3.2 Routed Port Configuration

Routers can have Ethernet (10 Mbps), Fast Ethernet (100 Mbps), Gigabit Ethernet (1,000 Mbps), and 10 gigabit (10 GB), Serial, and ATM interfaces. A router can have multiple interfaces, and the steps for configuring each interface are basically the same. Each interface is assigned a number. In our case, Cisco 2811 Router has 2 main slot on motherboard and 4 sub-slots on main slot: Slot 0,(0/0, 0/1, 0/2, 0/3) and Slot 1. The notation Serial0/0/0 indicates the

[interface-card-slot/port]. On Cisco's routers, a routed port can be configured simply by assigning an IP address to the interface. Once an IP address and its subnet mask are assigned to the interface and the interface is enabled, a Layer 3 network is created. The interface IP address becomes the gateway for that network. To program the interface, the router must be in the configuration mode. In the terminals, to enter privileged EXEC mode, or any other security level set by a system administrator, use the **enable** command. Entering privileged EXEC mode enables the use of privileged commands.

Click on R1 and CLI tab. It will prompt "**Would you like to enter the initial configuration dialog? [yes/no]:**" type "no" then press enter.



```
R1
Physical Config CLI Attributes
IOS Command Line Interface
62720K Bytes of NVRAM: System Configuration, 192416 bits of configuration.
Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 06:21 by pt_rel_team

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Willowbrook
Willowbrook(config)#
```

R1# enable

To access global configuration mode, use the **configure terminal** command in privileged EXEC mode:

R1# enable

R1# configure terminal

As shown below, note that the system prompt changes to indicate that you are now in global configuration mode.


```
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
```

To set the hostname, enter the command **hostname name**.

```
R1# enable
R1# configure terminal
R1(config)# hostname Willowbrook
```

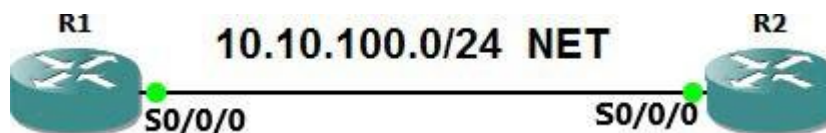
```
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname Willowbrook
Willowbrook(config)#
```

To select a particular Serial interface for configuration, use the **interface serial slot/port** command in global configuration mode.

```
R1# enable
R1# configure terminal
R1(config)# hostname Willowbrook
Willowbrook(config)# interface serial 0/0/0
```

Note that the system prompt changes to indicate the interface mode.

The next step is to assigning IP addresses to the network interface. An IP address identifies a location to which IP datagrams can be sent. To set a primary IP address for an interface, we use the command **ip address ip- address mask**.



```
R1# enable
R1# configure terminal
R1(config)# hostname Willowbrook
Willowbrook(config)# interface serial 0/0/0
Willowbrook(config-if)# ip address 10.10.100.1 255.255.255.0
```

Use the **shutdown** interface configuration command to disable an interface. Use the **no** form of this command to restart a disabled interface.

```
R1# enable
R1# configure terminal
R1(config)# hostname Willowbrook
Willowbrook(config)# interface serial 0/0/0
Willowbrook(config-if)# ip address 10.10.100.1 255.255.255.0
Willowbrook(config-if)# no shutdown
```

Notice that the router prompts you that the line protocol on interface Serial 0/0/0 changed state to down because other side of link which is R2 is not configured yet.

```
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Willowbrook
Willowbrook(config)#interface serial 0/0/0
Willowbrook(config-if)#ip address 10.10.100.1 255.255.255.0
Willowbrook(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Willowbrook(config-if)#
```

To exit any configuration mode to the next highest mode in the Command-Line Interface (CLI) mode hierarchy, use the **exit** command in any configuration mode.

```
R1# enable
R1# configure terminal
Router(config)# hostname Willowbrook
Willowbrook(config)# interface serial 0/0/0
Willowbrook(config-if)# ip address 10.10.100.1 255.255.255.0
Willowbrook(config-if)# no shutdown
Willowbrook(config-if)# exit
```



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Willowbrook
Willowbrook(config)#interface serial 0/0/0
Willowbrook(config-if)#ip address 10.10.100.1 255.255.255.0
Willowbrook(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Willowbrook(config-if)#exit
Willowbrook(config)#
```

To end the current configuration session and return to privileged EXEC mode, use the **end** global configuration command.

```
R1# enable
R1# configure terminal
R1(config)# hostname Willowbrook
Willowbrook(config)# interface serial 0/0/0
Willowbrook(config-if)# ip address 10.10.100.1 255.255.255.0
Willowbrook(config-if)# no shutdown
Willowbrook(config-if)# exit
Willowbrook(config)# end
```

Note that the system prompt changes to indicate that you exit from the global configuration mode.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Willowbrook
Willowbrook(config)#interface serial 0/0/0
Willowbrook(config-if)#ip address 10.10.100.1 255.255.255.0
Willowbrook(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Willowbrook(config-if)#exit
Willowbrook(config)#end
Willowbrook#
%SYS-5-CONFIG_I: Configured from console by console
Willowbrook#
```

To verify the status of the router interfaces, we can use the command **show ip interface brief** at the prompt (RouterName#).

```
R1# enable
R1# configure terminal
R1(config)# hostname Willowbrook
Willowbrook(config)# interface serial 0/0/0
Willowbrook(config-if)# ip address 10.10.100.1 255.255.255.0
Willowbrook(config-if)# no shutdown
Willowbrook(config-if)# exit
Willowbrook(config)# end
Willowbrook# show ip interface brief
```

```
Willowbrook#show ip interface brief
Interface                IP-Address      OK? Method Status
Protocol
FastEthernet0/0          unassigned      YES unset  administratively down down
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/0/0              10.10.100.1     YES manual    down
Serial0/0/1              unassigned      YES unset  administratively down down
FastEthernet1/0          unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Willowbrook#
```

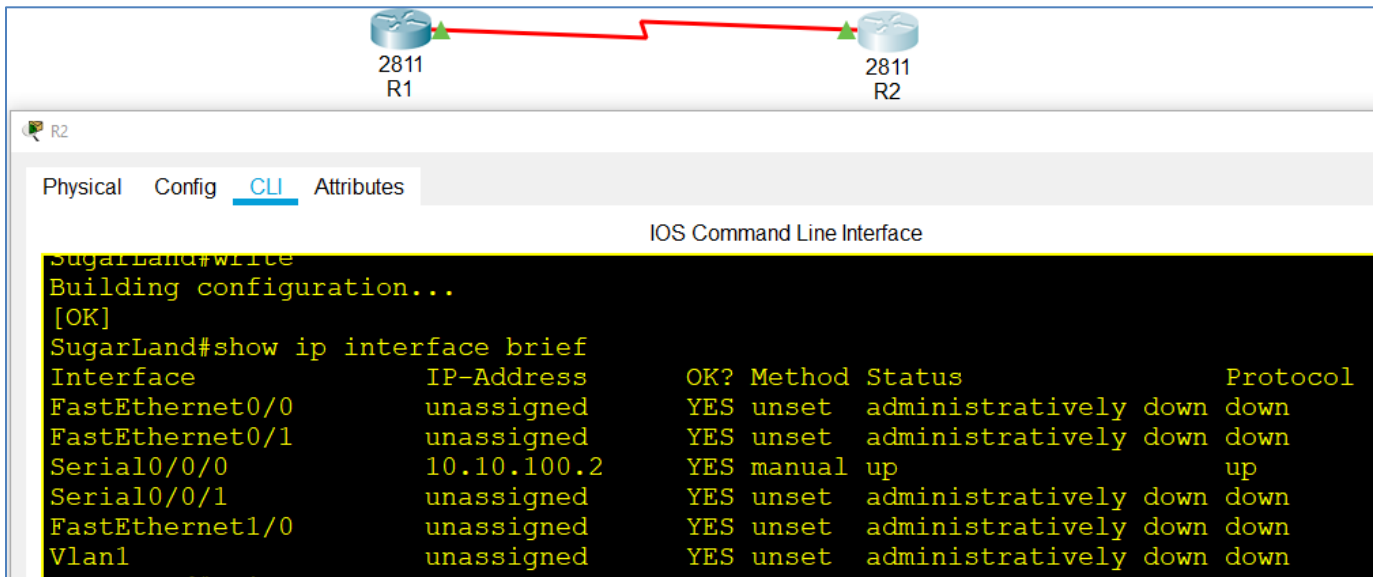
To save your configuration use **write** command. Otherwise, all your configuration will be lost after reboot.

```
Willowbrook#show ip interface brief
Interface                IP-Address      OK? Method Status
Protocol
FastEthernet0/0          unassigned      YES unset  administratively down down
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/0/0              10.10.100.1     YES manual    down
Serial0/0/1              unassigned      YES unset  administratively down down
FastEthernet1/0          unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Willowbrook#write
Building configuration...
[OK]
Willowbrook#
```

Please repeat all the steps for the **second router R2 except hostname should be SugarLand and IP address 10.10.100.2 255.255.255.0.**

```
R2# enable
R2# configure terminal
R2(config)# hostname SugarLand
SugarLand(config)# interface serial 0/0/0
SugarLand(config-if)# ip address 10.10.100.2 255.255.255.0
```

```
SugarLand(config-if)# no shutdown
SugarLand(config-if)# exit
SugarLand(config)# end
SugarLand# show ip interface brief
```



Now both router's link should be in green color as shown above. Both router's Serial0/0/0 Status and Protocol must be up.

3.3 Test Connectivity

The ping (Packet InterNet Groper) command is a very common method for troubleshooting the accessibility of devices. It uses two Internet Control Message Protocol (ICMP) query messages, ICMP echo requests, and ICMP echo replies to determine whether a remote host is active. The ping command also measures the amount of time it takes to receive the echo reply. The ping command first sends an echo request packet to an address, and then it waits for a reply.

In the console window for **SugarLand** router, please test if **SugarLand** router is able to ping **Willowbrook** router.

```
SugarLand#ping 10.10.100.1
```

```
SugarLand#ping 10.10.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 60/64/76 ms
SugarLand#
```

Repeat the same steps for the **Willowbrook** router. In the console window for **Willowbrook** router, please test if **Willowbrook** router is able to ping **SugarLand** router.

```
Willowbrook#ping 10.10.100.2
```

```
Willowbrook>enable
Willowbrook#ping 10.10.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/8 ms
Willowbrook#
```

Please do not forget to save your Cisco Packet Tracer file.

Congratulations! You have completed the Lab part I.

4. References

- [1] http://www.cisco.com/public/scc/compass/3600/information_files/inf_3600_interface_numbering.htm#fig_3
 - [2] http://www.cisco.com/en/US/products/hw/routers/ps274/products_data_sheet09186a0080091b89.html
 - [3] <http://www.cablesandkits.com/cisco-1port-fast-ethernet-network-module-nm1fetx-p-867.html>
 - [4] A Practical Guide to Advanced Networking, 3rd Edition
 - [5] http://www.cisco.com/c/en/us/td/docs/ios/12_2/configfun/command/reference/ffun_r/frf001.html#wp1017386
 - [6] http://www.cisco.com/c/en/us/td/docs/ios/12_2/configfun/configuration/guide/ffun_c/fcf001.html
 - [7] http://www.cisco.com/c/en/us/td/docs/security/asa/asa90/configuration/guide/asa_90_cli_config/basic_hostname_pw.html
 - [8] http://www.cisco.com/c/en/us/td/docs/ios/12_2/ip/configuration/guide/fipr_c/1cfipadr.html
 - [9] http://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960/software/release/12-2_55_se/command/reference/2960_cr/cli3.html
- <http://www.cisco.com/c/en/us/support/docs/ip/routing-information-protocol-rip/13730-ext-ping-trace.html>